Course Information

Course Code	MTH34501	Course Title	Deep Generative Models: A Statistical Perspective
Year/Semester	2026/ 2nd Semester	School	Department of Mathematical Sciences
Class Time/Classroom	TUE THU 14:30 – 15:45 (108-318(G))	Grading Type	Letter grade

Instructor Information

Instructor	Office	Tel.	E-mail	Office Hours
Younggeun Kim 김영근	108-301-11		Younggeun91 @unist.ac.kr	MON 14:00-15:00 TUE 16:00-17:00 Or by appointment

Teaching Method

Offline

Course Objectives & Description

Course Objectives: This course covers the mathematical foundations of generative models, ranging from classical frameworks to early deep architecture. It provides the essential background knowledge required for advanced coursework in generative models. Key topics include the foundations of probabilistic modelling, likelihood ratio-based generative models, and generative models utilizing alternative statistical distances.

Description: This course provides essential groundwork for advanced courses such as AI518: Deep Generative Models. Due to time constraints, the lectures will focus primarily on the theoretical foundations. However, please feel free to use office hours to ask any questions

about generative models, including implementation, coding aspects, and research topics.

There is no required textbook for this course. The lecture notes are designed to be self-contained, covering all necessary course content, though the references listed in Textbook & Reference Section may serve as helpful supplementary materials. Useful resources also include [1], which is a short pilot lecture series for this course that includes pilot video recordings and lecture notes, and [2], which was Ph.D.-level graduate course offered at Michigan State University and taught by Dr. Younggeun Kim.

Given that deep generative models have emerged relatively recently, to the best of the instructor's knowledge, there is currently no undergraduate-level course other than this one that covers everything from the basics of probabilistic modeling to (early) deep learning approaches. I would be grateful if you could share any relevant course materials or information you might find.

Prerequisites: A Student Background Survey (Extra Credit [EC] 1%) will be conducted to better accommodate students' diverse backgrounds.

Given that this is a 300-level course, students are assumed to be familiar with relevant foundational 100-level courses, such as Calculus I (MTH111), Calculus II (MTH112), and Introduction to AI Programming I (ITP107). Although there is no formal prerequisite course for this class, students are expected to be familiar with basic linear algebra and probability theory covered in relevant courses, such as Applied Linear Algebra (MTH203), Linear Algebra (MTH204), Mathematical Foundations of Machine Learning (MTH210), Statistics (MTH211), and Mathematical Analysis I (MTH251).

Such preliminary knowledge will be briefly reviewed in the first few weeks, and students are encouraged to cover further topics independently. For those who need to catch up on further details, useful materials include [3] and [4].

Grading

Attendance (%)	Midterm Exam (%)	Final Exam (%)	Quizzes (%)	Individual Assignments (%)
10.0	30.0	30.0		10.0

Reports (%)	Presentations (%)	Other (%)	Total (100%)	
10.0	10.0		100.0	

Remarks

Attendance: If a student's number of absences exceeds the limit established by university policy, the final grade will be a failing grade (F). While exceptions may be made for valid reasons in accordance with university policy, provided that the instructor's approval is obtained in advance (except in urgent situations detailed by university policy), excuses such as forgetting to check attendance or mobile phone issues may not be accepted.

Homework Assignments: There are six homework assignments, and the lowest score will be dropped. Assignments consist mostly of exercises from the lecture notes. Some selected problems will be graded for accuracy, while others will be graded solely for completion. Hard copies are due at the end of class meeting, and no late submissions will be accepted.

Exams: There will be one midterm and one final exam. All exams are closed-note and will be taken in person in the classroom on the following dates:

Midterm (30%): 10/22, 14:30 – 16:30 (120 mins). Location (Tentative): 108-318(G)

Final (30%): 12/17, 14:30 – 16:30 (120 mins). Location (Tentative): 108-318(G)

If the final class size is suitable to hold exams in the classroom, we will hold them there. Otherwise, we may reserve a separate room for the exams and announce the location accordingly.

Projects: There is one group project and one personal project:

- Group project: Choose a topic on the broader impacts of generative models and conduct a case study. Example agendas include:
  i. Introducing AI into peer-reviewing processes (e.g., Pre-peer review screening [5])
  ii. Utilizing generative AI for psychotherapy (e.g., helping mourners cope with grief [6] and [7])

This project aims to provide students with exposure to broader applications of generative models rather than reviewing technical details of their implementation. Findings will be delivered via an oral presentation during class. Further details will be announced later.

- Personal project: Write a review of generative adversarial networks paper [8] or variational autoencoder paper [9] and a meta-review of their peer-reviewing processes [10, 11]. Further details will be announced later.

Make-up policy: There will be no make-up exams. If a student misses an exam for a valid reason in accordance with university policy, the weight of the missed one may be reallocated to the next scheduled ones or distributed among the remaining ones.

Course Evaluation: Attendance (10%), Six homework assignments (2% each; lowest score dropped, total 10%), one group project presentation (10%), one personal project report (10%), one midterm exam (30%), and one final exam (30%).

Extra credit (EC) will be available for completing the Student Background Survey (1% EC) and may also be offered for advanced problems on the midterm and final exams.

Weekly Schedule

The schedule and topics for each class meeting are tentative and subject to change depending on class size and the pace of the lectures.

Week	Contents
01	09/01: Chapter 1: Fundamentals of Probability Theory (Part I) 09/03: Chapter 1: Fundamentals of Probability Theory (Part II); Student Background Survey Released; HW1 Released
02	09/08: Chapter 2: Random Variables (Part I) 09/10: Chapter 2: Random Variables (Part II); **Student Background Survey Closes (1% EC)**; **HW1 Due**; HW2 Released
03	09/15: Chapter 2: Random Variables (Part III) 09/17: Chapter 3: Likelihood Maximization Principle (Part I)
04	09/22: Chapter 3: Likelihood Maximization Principle (Part II); **HW2 Due**; HW3

	Released **09/24: Holiday – Chuseok (Korean Thanksgiving Day)**
05	09/29: Chapter 4: Basics of Probabilistic Modelling (Part I) 10/01: Chapter 4: Basics of Probabilistic Modelling (Part II)
06	10/06: Chapter 4: Basics of Probabilistic Modelling (Part III) 10/08: Chapter 4: Basics of Probabilistic Modelling (Part IV); **HW3 Due**; HW4 Released
07	10/13: Chapter 4: Basics of Probabilistic Modelling (Part V) 10/15: **Group Project Presentation** * Depending on class size, we may also allocate presentation time on 10/13
08	10/20: Chapter 5: Autoregressive model 10/22: **Midterm** (Scope: Chap. 1, 2, 3 & 4)
09	10/27: Chapter 6: Energy-based Models 10/29: Chapter 7: Variational Autoencoders (Part I)
10	11/03: Chapter 7: Variational Autoencoders (Part II) 11/05: Chapter 7: Variational Autoencoders (Part III); **HW4 Due**; HW5 Released
11	11/10: Chapter 7: Variational Autoencoders (Part IV) 11/12: Chapter 8: Generative Adversarial Networks (Part I)
12	11/17: Chapter 8: Generative Adversarial Networks (Part II) 11/19: Chapter 8: Generative Adversarial Networks (Part III); **HW5 Due**; HW6 Released
13	11/24: Chapter 8: Generative Adversarial Networks (Part IV) 11/26: Chapter 9: Integral Probability Metrics-based Approaches
14	12/01: Chapter 10: Wasserstein Distances-based Approaches (Part I) 12/03: Chapter 10: Wasserstein Distances-based Approaches (Part II); **HW6 Due**
15	12/08: Chapter 11: Score-based Approaches (Part I) 12/10: Chapter 11: Score-based Approaches (Part II); **Personal Project Report Due**
16	12/17: **Final** (Scope: All course content)

Notes and Remarks

Textbook & Reference

Text/ Reference	Resource Type	Title	Author	Publisher	Year	Edition	ISBN/ISSN	Course Reserves Designation
Textbook	eBook	The elements of statistical learning: data mining, inference and prediction	Hastie, Trevor	Springer	2009	2nd	978-0-387-84857-0	
References	Article	Nips 2016 tutorial: Generative adversarial networks	Ian Goodfellow	arXiv ID: arXiv: 1701.00.00160	2016	NA		

[1] *Deep Generative Model: A Statistical Perspective*; https://github.com/kyg0910/DeepGenerativeModel-A-Statistical-Perspective

[2] *STT997: Deep Generative Model: A Statistical Perspective*; https://github.com/kyg0910/-STT997-Deep-Generative-Model/tree/main

[3] Ross, Sheldon M. A first course in probability. Vol. 8. London: Pearson, 2014.

[4] Rice, John A. *Mathematical statistics and data analysis*. Vol. 9. No. 3. Belmont, CA: Thomson brooks/cole, 2007.

[5] Checco, Alessandro, et al. *AI-assisted peer review*. Humanities and Social Sciences Communications 8.1 (2021): 1-11.

[6] MBClife, *[VR Human Documentary] Mother meets her deceased daughter through VR technology* https://www.youtube.com/watch?v=uflTK8c4w0c, February 2020. Accessed: 2026-08-31

[7] CBS News, *AI simulations of loved ones help some mourners cope with grief*. https://www.cbsnews.com/news/ai-grief-bots-legacy-technology/, June 2024. Accessed: 2026-08-31

[8] Goodfellow, Ian J., et al. "Generative adversarial nets." Advances in neural information processing systems 27 (2014).

[9] Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." arXiv preprint arXiv:1312.6114 (2013).

[10] NeurIPS 2014. Reviews for the submission ``Generative Adversarial Nets'' https://proceedings.neurips.cc/paper_files/paper/2014/file/f033ed80deb0234979a61f95710dbe25-Reviews.html, December 2014. Accessed: 2026-08-31

[11] ICLR 2014. Reviews for the submission ``Auto-Encoding Variational Bayes'' https://openreview.net/forum?id=33X9fd2-9FyZd, December 2013. Accessed: 2026-08-31